

Examiner reports

Q1.

Part (a) was generally correct. $\frac{e^0}{6}$ was not good enough and neither was 0.16. Wrong answers included 1 and $\ln 6$.

In part (b) many solutions were correct, often in an unsimplified form, but $\frac{1}{6x}$ and $\frac{6}{x}$ were common. Those who included dx or $+c$ in their solution were penalised.

Q2.

In part (a)(i) the majority of candidates were able to differentiate e^{2x} and apply the product rule correctly but a few made an error in one or other of these processes. Occasionally an attempt to simplify was incorrect.

In (a)(ii) when finding the equation of the tangent (a line) it is essential to find the gradient (a constant) at the requisite point first. Most did this but it is disconcerting, at this level, to find candidates who give a non-linear equation as their answer by failing to do the first step. A few added $e^2 + 2e^2$ wrongly or lost exactness by evaluating $3e^2$, and a few found the equation of the normal.

In (b) although the quotient rule was known by almost everyone, and almost all earned the first mark, some appalling algebra and trigonometry abounded thereafter. A few failed even to get the first mark as they put a term in the wrong place on the numerator, had $\sin x$ or $\cos x$ instead of $\sin 3x$ or $\cos 3x$ in one of their differentials or else their denominator was $1 + \cos^2 3x$ instead of

$(1 + \cos 3x)^2$. A few failed to use the chain rule when differentiating $\sin 3x$ or $\cos 3x$. Those who missed the brackets around $(1 + \cos 3x)$ in the numerator sometimes recovered but far too many falsely cancelled $\cos 3x$ at this stage. If the numerator was correctly expanded another mark was available, although $\cos 3x^2$ and $\cos 3^2 x$ instead of $\cos^2 3x$ for $(\cos 3x)^2$ was common; this was recoverable, but $\cos 9x^2$ was not. After that more incorrect cancellation was seen. Taking out a common factor of 6 was the safest route forward, as those who tried to deal with $6\cos^2 3x + 6\sin^2 3x$ often put it equal to 1 or 18 instead of 6. Even being correct to that stage did not ensure completion as, again there was false cancellation. Those who split the terms, then used $(1 - \cos^2 3x)$ for $\sin^2 3x$ and factorised to $(1 - \cos 3x)(1 + \cos 3x)$ generally completed successfully. It was quite extraordinary how many methods with completely wrong working

landed up at $\frac{6}{1 + \cos 3x}$.

Q3.

In part (a), almost all candidates were able to find the first and second derivative correctly, although there was an occasional arithmetic slip and some could not cope with the fraction term.

In part (b), those who substituted $t = 2$ into $\frac{dy}{dt}$ did not always explain that $\frac{dy}{dt} = 0$ is the condition for a stationary point. Some **assumed** that a stationary point occurred when $t = 2$, went straight to the test for maximum or minimum and only scored half the marks.

It was advisable to use the second derivative test here; those who considered values of $\frac{dy}{dt}$ on either side of $t = 2$ usually reached an incorrect conclusion because of the proximity of another stationary point.

In part (c)(i), the concept of 'rate of change' was not understood by many who failed to realise the need to substitute $t = 1$ into $\frac{dy}{dt}$. Some candidates wrongly substituted $t = 1$

into the initial expression for y or into their expression for $\frac{d^2y}{dt^2}$ and these candidates

were unable to score any marks at all on this part. Even those who used $\frac{dy}{dt}$ sometimes made careless arithmetic errors when adding three numbers.

In part (c)(ii), it was not enough to simply write the word "increasing": some explanation

about $\frac{dy}{dt}$ being positive was also required. Some candidates erroneously found the value of the second derivative when $t = 1$ or calculated the value of y on either side of $t = 1$.

Q4.

In part (a)(i), many candidates did not realise that differentiation was required in order to find the gradient of the curve, but instead erroneously used the coordinates of O and A . Some tried to rearrange the terms but usually made sign errors in doing so. In part (a)(ii), those who had the correct gradient in part (a)(i) were usually successful in finding the correct equation of the normal, though not everyone followed through to the required form, and sign errors were common. However, most obtained at least a method mark here, unless they found the equation of the tangent. The main casualties were once again those who always use the $y = mx + c$ form for the equation of a straight line.

In part (b)(i), most candidates were well drilled in integration and scored full marks, although some wrote down terms with incorrect denominators. The limits 2 and 0 were usually substituted correctly, but it was incredible how many could not evaluate $32 - 38 - 8$ without a calculator. The correct value of the integral was -14 , but far too many thought that they had to change their answer to $+14$ and so lost a mark. A small number of candidates differentiated or substituted into the expression for y rather than the integrated function.

In part (b)(ii), there was still some apparent confusion about area when a region lies below the x -axis. The area of the triangle was 6 units and hence the area of the shaded region was $14 - 6 = 8$ but, not surprisingly, there were all kinds of combinations of positive and negative quantities seen here. It was worrying to see so many candidates failing to calculate the triangle area, with several finding the length of OA instead. Some able

candidates found the equation of OA and the area under it by integration, but this was not the expected method.

Q5.

There were many correct answers for the derivative in part (a) although the incorrect answer $-3x^{-2} + 48$ was not a rarity. Candidates who wrote down $y = x^{-3} + 48x$ before differentiating generally scored more marks than those who just stated an answer. Many

candidates appreciated the need to put $\frac{dy}{dx} = 0$ in part (b) but poor algebraic manipulation or a failure to include '±' frequently resulted in low marks for this part of the question.

Even those who obtained the correct coordinates of the stationary points did not always appreciate that the required tangents had equations $y = \pm 32$. Although the marks for finding the equation of the normal in part (c) were generally better than for the work in part (b), it was surprising to see a significant minority giving the equation of the tangent at the given point.

Q6.

Part (a) was well answered by the majority of candidates. Many fully correct responses were seen and, if there were errors, it was usually the final accuracy mark that was lost through the omission of brackets at some stage in the solution.

Part (b) was not answered as well as part (a). Although many candidates were successful in factorising the required quadratic, there were also many solutions which were accompanied by terms in e^{-4x} . Where marks were lost, it was mainly due to incorrect signs, though some candidates did manage to obtain follow through marks.

A few candidates attempted to substitute their values of x back into the derivative rather than y . Many candidates stopped when they had found the two values of x , presumably thinking they had finished the question.

Q7.

In part (a)(i), most candidates stated the formula for $\sin 2\theta$ correctly although not always in its simplest form.

In part (a)(ii), although many candidates knew the formula for $\cos 2\theta$ in at least one its correct forms, several gave $\cos^2 \theta + \sin^2 \theta$, and later in the question just gave the value of $\cos 2\theta$ as 1 without apparently thinking something could be wrong.

In part (a)(ii), many candidates had problems. It appeared that the 3-4-5 triangle was either not well known or not recognised as relevant; although some candidates did draw this to justify their answer others used $\sin^2 \theta = 1 - \cos^2 \theta$. Many candidates worked backwards from the answer given, to find a value for $\sin \theta$, which gained no credit nor did going via an angle from the given value of $\cos \theta$. Many candidates omitted to find $\cos 2\theta$ or, as above, gave the value as 1, or another incorrect value from an incorrect expression. Those who did find the correct value for $\cos 2\theta$ often dropped the minus sign in later working.

In part (b)(i), the use of the chain rule with these parametric equations was generally done well with most candidates completing it correctly. Some candidates apparently tried to integrate rather than differentiate and some made sign errors when differentiating. A few candidates had the chain rule expression the wrong way up.

In part (b)(ii), although some neat and concise correct answers were seen, many candidates got in a mess trying to use angles and not working in exact fractions. Some confused the value of the sine and cosines involved with the angles writing terms such as $\sin \frac{3}{5}$. Some left a mix of fractions and decimal numbers in their final answer, which is bad practice but was accepted if the values of the coordinates at P were correct.

Q8.

Part (a)(i) Most candidates were able to integrate the expression with only the weakest candidates unable to do this basic integration. Poor notation was used with many including the integral sign after integrating. It would have been thought that this bad habit would have been corrected by the time of the examination. Many candidates did not find the actual value of the definite integral until part (a)(ii) and on this occasion full credit was given. It was alarming that many candidates who had correct fractions were unable to combine these to give a value of 8.4 or equivalent. Weaker candidates were seen

substituting values into the expression for y or $\frac{dy}{dx}$, showing a complete lack of understanding.

Part (a)(ii) It was necessary to consider a rectangle of area 18 and then to subtract their answer from part (a)(i) in order to obtain the area of the shaded region. Many believed that the area of the rectangle was 9 and others failed to do this basic subtraction correctly, even when their answer to part (a)(i) was correct.

Part (b)(i) Many candidates did not realise the need to find $\frac{dy}{dx}$ before substituting the value $x = 1$ and thus failed to score some easy marks for finding the gradient of the curve. A substantial number of candidates tried to calculate the gradient of the straight line between two points on the curve and scored no marks for this.

Part (b)(ii) Unfortunately many candidates tried to find the equation of the normal instead of the tangent to the curve. Otherwise, since there was a generous follow through in this part of the question, most were able to score this final mark. The only exceptions were those who insisted on using $y = mx + c$ where poor arithmetic often prevented them from finding a value for c .

Q9.

Part (a)(i) Usually after a few abortive attempts many candidates realised that they had to add together the areas of the various faces. Once they had the correct expression for the total surface area most candidates were able to obtain the printed result. There was clearly some fudging on the part of weaker candidates and they could earn little more than a single method mark.

Part (a)(ii) This was surprisingly one of the biggest discriminators on the paper with only the best candidates being able to obtain the correct expression for V . Trying to make y the

subject of the equation from part (a)(i) caused problems for many who took this approach; others substituted for xy in the formula $V = 6x(xy)$ and were often more successful. Once again it was quite common to see totally incorrect expressions being miraculously transformed into the printed answer.

Part (b)(i) Most candidates scored full marks for this basic differentiation although it was

not always clearly identified as $\frac{dV}{dx}$ in their working.

Part (b)(ii) Most substituted $x = 2$ into their expression for $\frac{dV}{dx}$ and found the value to be zero. It was then necessary to make a statement about the implication of there being a stationary value in order to score full marks.

Part (c) Some made a sign error when finding the second derivative, but the majority of candidates scored full marks in this part. Credit was given if the correct conclusion was drawn from the sign of their second derivative, provided no further arithmetic errors occurred.

Q10.

Most candidates coped with the negative fractional powers well, differentiated correctly and found an equation of a normal, although there were a few equations of tangents presented as final answers to part (b)(ii). A common arithmetical slip in part (b)(ii) was

$\frac{1 + \sqrt{1}}{1}$, incorrectly evaluated as 1. The expression for the second derivative was often correct in part (c)(i).

Part (c)(ii) was discriminating. The most popular approaches were to either put $x = 1$ in

$\frac{d^2 y}{dx^2}$, or to equate $\frac{d^2 y}{dx^2}$ to zero and solve, for which no marks were scored.

A significant number of candidates did, however, earn some credit by making a more general comment on the sign of the second derivative but many lacked rigour and, in particular, assumed the second derivative has to be negative for a maximum, overlooking the possibility that the second derivative may be zero at a maximum.

Q11.

- (a) (i) The first part of this question was reasonably answered with many candidates obtaining both marks. Where candidates obtained 1 mark it was because many ended up with the answer $1/(5x - 2)$.
- (ii) Again this part was very well answered with candidates obtaining both marks. The majority of candidates arrived at $K \cos 2x$ but $K = -2$ was a common error.
- (b) (i) This part was not answered very well. Many candidates lost a mark through using -0.693 instead of $\ln\left(\frac{1}{2}\right)$ and $f(x) \geq 0$ was a common response.
- (ii) Most candidates were able to do this part with the correct answers often seen.

The main error was the omission of brackets around $5x - 2$ obtaining $\sin^2(\ln 5x - 2)$. The expression for $fg(x)$ was also often seen.

- (iii) For those candidates with a correct starting expression many went on to get full marks. Those candidates who used $\sin \ln (5x - 2)^2$ often lost an accuracy mark for not rejecting one of their answers. Most candidates obtained the first method mark for making the correct initial step for their expression.
- (iv) This was usually well done but there were common errors of dividing by $\sin^2$ obtaining $y/\sin^2 = x$ or even $y/\sin = 2x$ and $y/2 = \sin x$

Q12.

Many candidates lost marks on this question from careless work when handling the number of e^{2x} terms involved.

- (a) Most candidates obtained the correct coordinates although $(\ln x, 0)$ was a common error.
- (b) This was generally well answered by those candidates able to use the quotient rule. Some errors with the numerator being reversed to $(\ln x - 1)$ meant candidates lost accuracy marks later in the question. Many candidates obtained 3 marks for reaching $\ln x - 1 = 0$ and many went on to find $x = e$. However many stopped here without finding the value of y .
- (c) Although many correct solutions were seen some candidates lost marks through leaving $(e^{3x})^2$ rather than simplifying it. There was also a large number of candidates who did not answer part (b) but started again here, used the quotient rule correctly and went on to then answer the remainder of the question correctly. They seemed to realise that they needed the derivative to find a gradient but not for finding stationary points

Q13.

- (a) The vast majority of candidates completed this question successfully and as expected, by finding the two derivatives with respect to t and combining them using the chain rule. The more common errors were to omit the squared powered from $6t^2$ or the negative sign in -3 . Some candidates lost a mark through not simplifying their final result. Relatively few candidates made the error of multiplying the derivatives or having the chain rule upside down. Very few candidates took the approach of eliminating t and seeking an equation $y = f(x)$. Of those who did, hardly any were successful in differentiating their expression.
- (b) The vast majority of candidates was successful here too, substituting $t = 1$ immediately and obtaining the gradient of the normal correctly and then a correct form of the equation of the normal. Those few candidates who made no attempt to find the gradient of the normal and just used the tangent gained no credit. The fairly rare error in finding the gradient of the normal was to omit the negative sign. Some candidates also failed to substitute for the given value for t and gained no credit.
- (c) Various correct approaches were seen with the most efficient being to obtain t in terms of x and substitute into $y = f(t)$, and many candidates did this correctly. The

common errors were to omit or have an unclearly positioned negative sign, or to make an error in the power. Other attempts at elimination were generally not so successful. Many candidates wasted time through proceeding to expand their cartesian equation, or just generally and unnecessarily manipulating it algebraically. Once t is eliminated, the resulting equation is a cartesian equation.

Q14.

In part (a) almost all candidates were able to find the first and second derivative correctly, although there was an occasional arithmetic slip and some could not cope with the fraction term.

Those who substituted $t = 3$ into $\frac{dx}{dt}$ in part (b) did not always explain that $\frac{dx}{dt} = 0$ is the condition for a stationary point. Some assumed that a stationary point occurred when $t = 3$ and went straight to the test for maximum or minimum and only scored half the marks. It

was advisable to use the second derivative test here; those who considered values of $\frac{dx}{dt}$ on either side of $t = 3$ usually reached an incorrect conclusion because of the proximity of another stationary point.

In part (c) the concept of “rate of change” was not understood by many. Approximately

equal numbers of candidates substituted $t = 1$ into the expression for $\frac{dx}{dt}$ or $\frac{d^2x}{dt^2}$ and so only about half of the candidates were able to score any marks on this part. Those who used $\frac{dx}{dt}$ often made careless arithmetic errors when adding three numbers.

In part (d), as in part (c), candidates did not realise which expression to use and many wrongly selected the second derivative. It is a general weakness that candidates do not realise that the sign of the **first** derivative indicates whether a function is increasing or decreasing at a particular point.

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Q15.

In part (a)(i) the majority of candidates realised the need to find the value of $f(x)$ when $x = 2$. However, it was also necessary, after showing that $f(2) = 0$, to write a statement that the zero value implied that $x - 2$ was a factor. It was good to see more candidates being aware of this.

In part (a)(ii), those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

In part (b)(i) a surprising number of candidates failed to see the need to differentiate in order to find the gradient at Q . Those who attempted to find $\frac{dy}{dx}$ sometimes wrote it as $3x^2 + x$, but usually were aware of the need to substitute $x = 2$.

In part (b)(ii) those who had the correct gradient in part (b)(i) were usually successful in finding the correct equation of the tangent, and most obtained at least a method mark

here.

In part (b)(iii) most were well drilled in integration and earned full marks, although some

wrote $\frac{x^4}{4} + \frac{x^2}{2} - 10$ and others gave $\frac{x^4}{4} + \frac{x^2}{2} - \frac{10^2}{2}$ as their answer. For part (b)(iv) the correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 0 and 2 and it was incredible how many could not evaluate $6 - 20$ without a calculator. Very few candidates realised the need to show clearly that, although the integral from 0 to 2 gave a value of -14 , the area of the shaded region was 14. A separate statement was needed and those who simply wrote $4 + 2 - 20 = -14 = 14$ did not score full marks. Those more able candidates who made a statement about the region being entirely below the x -axis and who subsequently evaluated the integral from 2 to 0 correctly scored full marks.

Q16.

This long calculus question with a final part covering transformations was a relatively high source of marks for many candidates. Most candidates were able to correctly differentiate

$2x^{\frac{3}{2}}$, but not all evaluated it correctly at $x = 4$. The majority of candidates showed a good understanding of the method involved in finding the equation of the normal in part (a)(ii), although some others did not use the correct gradient. Although the vast majority of

candidates knew the rules for integrating $8x^{\frac{1}{2}}$, there was a significant number of such candidates who did not simplify the coefficient correctly.

It was not uncommon to see either ' $\frac{8}{3}x^{\frac{3}{2}}$ ', not simplified or simplified incorrectly to ' $12x^{\frac{3}{2}}$ '.

Part (b)(ii) was generally answered better than expected, with a majority of candidates applying a correct method for finding the area of the shaded region. Although many candidates realised that the transformation was a 'translation', there were frequently sign errors in the vector, or components were interchanged. It was also disappointing to see a significant increase in the number of descriptions that involved more than one transformation which, as usual, was given no marks; the two transformations quoted were translation and a stretch.

Q17.

There was a significant improvement from June 2008 in candidates' solutions to the binomial expansion topic. Part (a) was well answered, although a lack of brackets continues to be the main source of errors. Many candidates illustrated that they knew how to use their answer to part (a) to write down the expansion of $(1 - 2x)^4$, but not all solutions which finished with the printed answer were entirely convincing.

Part (c) was not always attempted and, even when attempted, the majority failed to convincingly show that there was only one stationary point. Many solutions included the correct equation ' $96x + 128x^3 = 0$ ', but a significant proportion of these lost the ' $x = 0$ ' root

and, by erroneously changing a sign, ended up with a value for x of the form $\frac{\sqrt{3}}{2}$.

Q18.

In part (a), most candidates used the product rule correctly, but at this stage $e^{2x} \neq 0$ was rarely seen, although many candidates obtained the correct quadratic equation. Factorisation, or use of the formula, was usually done well and many candidates obtained both of the correct solutions. Where candidates seemed unsure of what to do with the e^{2x} , many expressions involving the use of $\ln e^{2x} = 2x$ were seen together with the formation of a cubic expression in x . A third value of $x = 0$ was also seen by candidates who thought this was the solution of $e^{2x} = 0$.

Part (b) was generally well answered by those candidates able to do the product rule in part (a). Some errors with the coefficients meant candidates lost accuracy marks.

In part (c), although many correct solutions were seen some candidates lost marks

through incorrect evaluation of $\frac{d^2 y}{dx^2}$.

Q19.

Candidates were generally successful in part (a), sometimes aided by the allowances made for $(-1, 180)$.

Part (b) was well answered by many candidates. Common errors were the use of 'shift' and 'transformation' to describe the word 'translation', and the stretch was often given a scale factor of $\frac{1}{2}$ in the x -axis.

In part (c), the shape of the graph was usually correct, but many candidates lost this mark by going into other quadrants. The mark for the labelling of the axis was not so readily earned, as these values were often missing or incorrect.

In part (d)(i), many candidates gave the correct response. The major error was to state that $2y = \cos^{-1}(x - 1)$ for the first step resulting in no marks being awarded.

In part (d)(ii), most candidates at least obtained the method mark. Where candidates correctly differentiated, they went on to achieve full marks.

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Q20.

Very few candidates achieved completely correct solutions to this question.

Part (a) was well answered by the majority of candidates. The main error was the two terms in the numerator being interchanged, resulting in $k = +12$. Other errors came from the expansion of the brackets, with various numerators including $8x$, and with -3 not being multiplied by 4, giving a final constant of -3 .

In part (b)(i), the expression $\ln(4x - 3)$ was rarely differentiated completely correctly. A

common error was to get $\frac{x}{4x-3} + \ln(4x-3)$, rather than $\frac{4x}{4x-3} + \ln(4x-3)$, which earned two method marks.

In part (b)(ii), $\frac{dy}{dx}$ was frequently calculated correctly, following through from part (b)(i),

but then this value was used as the gradient of the normal, with some candidates explicitly stating that they were finding the normal.

In part (c)(i), most of the candidates earned a method mark for ' $du = 4dx$ '. Many of these candidates went on to write the integral completely in terms of u . Some candidates lost

marks at this stage through either obtaining an extra factor of 4 or getting $x = \frac{u-3}{4}$ when rearranging $u = 4x - 3$.

Although many fully correct responses were seen, some candidates were unable to proceed beyond this point: part (c)(ii) proved to be beyond many candidates, as the correct application of integration by parts was not common. Most candidates started the

question with $u = \ln$ and $\frac{dv}{dx} = 4x - 3$. Those candidates who correctly started this part of the question often failed to get the first accuracy mark because of their derivative of $\ln(4x - 3)$, losing the factor of 4 in the numerator. However, many were able to get the next method mark because they correctly used their value from part (c)(i). A few candidates spotted that the result obtained in part (b)(i) could provide an alternative method of solution.

Q21.

In part (a) almost everyone obtained the correct expression for $\frac{dy}{dx}$, although a few spoiled their solution by dividing each term by 5 or adding "+ c" to their answer.

In part (b) most candidates substituted $x = -2$ into their expression for $\frac{dy}{dx}$, but, in order to score full marks, it was necessary to show $(-2)^4$ written as 16 or to show that $\frac{dy}{dx} = 80 - 80 = 0$ and then to write an appropriate conclusion about P being a stationary point.

For part (c) many candidates simply wrote down an expression for $\frac{d^2y}{dx^2}$ in terms of x when answering part (i) and only evaluated the second derivative when determining the nature of the stationary point in part (ii). On this occasion full credit was given, but candidates need to realise what is meant by the demand to "find the value of" since this may be penalized in future examinations.

In part (d) some candidates failed to find the y -coordinate of P , which was necessary in order to find the equation of the tangent. It was pleasing to see most candidates using the

value of $\frac{dy}{dx}$ when $x = 1$, but unfortunately many tried to find the equation of the normal instead of the tangent to the curve.

Q22.

In part (a) most candidates differentiated the equation of the curve correctly and the

majority of these recognised that at the maximum point the value of $\frac{dy}{dx}$ was 0, although some worked with $\frac{d^2y}{dx^2} = 0$.

In part (b) many candidates had difficulty solving the equation $22.5x^{\frac{1}{2}} - 2.5x^{\frac{3}{2}} = 0$. There was some invalid squaring of expressions and terms and also it was not uncommon to see $x^{\frac{3}{2}} \div x^{\frac{1}{2}}$ simplified incorrectly either to x^3 or to x^2 .

With the equation of the tangent printed in part (d) candidates had to convince examiners that they were finding the value of $\frac{dy}{dx}$ at $x = 1$ for the gradient of the tangent in part (c). Although not all candidates did this convincingly, the majority did show sufficient evidence to reach the printed equation.

The final part of the question proved to be a good discriminator at the top grade level. Although some excellent solutions were seen these were rare. The most common wrong method was to assume that the length of RM was either equal to the length of PM or equal to half the length of PM . Unfortunately a minority of candidates who produced better solutions which started with the correct equation ' $162 = 20x - 6$ ' failed to score the final mark due to an inability to solve this equation correctly.

Q23.

Part (a)(i) was not very well answered by the majority of candidates. Fully correct responses were seen but it was usually from candidates who successfully rearranged the equation into the form $f(x) = 0$, which was seen in a number of acceptable forms. Where candidates simply substituted in the two values given into the LHS they obtained 1 and 0 but still indicated that this was a change of sign and therefore there was a root. Most candidates achieved at least one mark but lost the second by simply stating change of sign therefore a root.

Very few incorrect responses were seen in part (a)(ii).

Part (a)(iii) was very well answered. The main error was with the candidates who used degrees rather than radians

Part (b)(i) was very well answered, with most candidates successfully using the quotient rule. Where errors occurred it was usually through missing brackets, although many candidates were able to recover at some stage in the working.

As part (b)(ii) followed on from part (i) most candidates were able to obtain the mark for substitution; even those who had incorrectly simplified their work. Many fully correct responses were seen. Several candidates also stopped after the substitution of $x = 0$ and left their answer as -2 .

Q24.

Most candidates lost marks on this question. Full marks were not common although they

were seen from the more able candidates.

Not many candidates obtained full marks for part (a). Most candidates scored 2 out of 4 since they started correctly by differentiating $\ln t$ and integrating $(t - 1)$. Obvious errors were differentiating both terms or starting by trying to integrate $\ln t$. Several candidates managed to obtain the second accuracy (A) mark for simplification to $(t^2/2 - t) \ln t - \int (t/2 - 1)dt$ but lost the final accuracy mark by ending up with a final term of $-t$, not $+t$.

Part (b) was done well by most candidates, although poor manipulation often cost the loss of the final accuracy mark. Some candidates also confused the issue by trying to introduce terms in u and du .

Part (c) was not very well answered, with candidates failing to change the limits as a common error. Many candidates did not appreciate that they should be using their answer to part (a) and tried to start again, often obtaining different answers to those they had found in part (a). After correctly approaching the choice of u and dv/dt , most were then defeated by the required manipulation of the subsequent algebra.

Q25.

In part (a), most candidates were able to find the correct expression for $\frac{dy}{dx}$, although there were some who left $+5$ in their answer or added $+C$.

In part (b), It had been expected that candidates would solve the equation $\frac{dy}{dx} = 0$ and obtain the equation $x^3 = 8$ and hence deduce that $x = 2$. It seemed, however, that many were unable to formulate an appropriate equation, but merely spotted the correct answer: $x = 2$. This was not penalised on this occasion, provided that the candidate stated clearly that the x -coordinate of M was equal to 2.

In part (c)(i), the expression for $\frac{d^2y}{dx^2}$ was usually correct.

In part (c)(ii), although the method was left open, most candidates found the value of the second derivative when $x = 2$ and correctly concluded that M was a minimum point.

In part (c)(iii), some candidates were not aware of the need to find the value of $\frac{dy}{dx}$ when $x = 0$ in order to ascertain whether the curve was increasing or decreasing at that point.

Q26.

In part (a)(i), a few candidates ignored the request to use the factor theorem and scored no marks for using long division. It was necessary to make a statement that " $x + 1$ is a factor", after showing that $f(-1) = 0$, in order to score full marks.

Part (a)(ii) was not answered as well as similar questions in previous years. Perhaps the sketch lured some into trying to write down three factors without any further working,

rather than using the intermediate step of showing that $p(x) = (x + 1)(x^2 - x - 6)$ before writing $p(x)$ as a product of three factors. Many who tried long division were flummoxed by there being no x^2 term.

In part (b)(i), those who had the correct linear factors in part (a)(ii) usually wrote down correctly that A had coordinates $(-2, 0)$, although some carelessly wrote the point as $(0, -2)$.

Many candidates simply found an indefinite integral in part (b)(ii) and then a definite integral in part (b)(iii). The two parts were generously treated holistically when candidates did this. The fractions once again caused problems to most candidates who are so used to having a calculator to do this work for them. It was very rare to see the correct answer of -32 for the definite integral.

In part (b)(iii), many lost out on an easy mark because they rolled their two sections into one: those who wrote “integral = $-32 = 32$ ” gained full credit for part (b)(ii) but did not score the mark in part (b)(iii). It was necessary to give a positive value for the area of the region and to make this explicit. In anticipation of a lot of wrong answers in part (b)(ii), a follow through mark was awarded in part (b)(iii): for example, if a candidate’s answer in part (b)(ii) was -20 and they concluded that the area was 20 in part (b)(iii), they scored the mark.

In part (b)(iv), most candidates differentiated correctly, but quite a few thought that $3(-1)^2 - 7$ was equal to -10 and thus obtained the wrong gradient of the curve.

In part (b)(v), a large number of candidates found the correct equation of the normal but some still confused tangents and normals and consequently thought that the gradient of the normal was equal to -4 . It was quite common for weaker candidates to either negate their gradient or take the reciprocal but to fail to do both.

Q27.

Generally, most candidates scored relatively high marks for this long structured calculus question. Parts (a)(i), (ii) and (iii) were answered very well, although some used the gradient of the normal in part (a)(iii) as -2 and so in part (a)(iv) obtained a positive value for the y -coordinate of Q . A glance at the diagram should have indicated that such a value was incorrect.

Part (a)(v), which required candidates to find the x -coordinate of the maximum point, M , was generally not answered with any confidence. To score any marks for this part, candidates were required to go beyond just equating their answer for part (a)(i) to zero. Many candidates again incorrectly believed that for a maximum point the second derivative was zero. Candidates generally found the correct answer to the indefinite integral in part (b)(i), but a significant minority forgot to add the area of the triangle to their ‘area under the curve’ in part (b)(ii) of the question.

Q28.

Part (a)(i) was well answered by the majority of candidates. Many fully correct responses were seen, and, if there were errors, it was usually through further incorrect work or by the omission of brackets.

Very few incorrect responses were seen to part (a)(ii). Most candidates appeared to be

able to use the product rule successfully. Some errors with signs were made, leading to the loss of the accuracy mark.

Part (b) was very well answered.

Q29.

Some candidates did not attempt part (a). However, those who could square $9 - 3x$ usually earned the marks. A surprising number substituted for both x and y and most were unable to complete the solution successfully.

In part (b)(i), the differentiation was carried out very well by most candidates. Errors in the value of k occurred due to taking out a common factor either before or after differentiating, so values of k such as 3 and 9 were common. A few weak candidates did not recognise that they had to differentiate V here, and generally were unable to proceed through the rest of the question.

In part (b)(ii), the factorisation was usually carried out successfully, and candidates who had not found the correct value of k were still able to earn these marks.

The most common error in part (c) was differentiating $x^2 - 4x + 3$, rather than $27(x^2 - 4x + 3)$, and this approach earned a method mark only.

Credit was given in part (d)(i) for substituting their values of x into their second derivative and so most candidates earned this mark, provided they had two values of x .

In part (d)(ii), most realised that $x = 1$ needed to be selected since $\frac{d^2y}{dx^2}$ was negative when $x = 1$.

In part (d)(iii), not all substituted $x = 1$ into the equation for V . Those who did often made arithmetic errors. Some gave the value of y when $x = 1$ and others gave the answer as

-54 , the value of $\frac{d^2y}{dx^2}$ when $x = 1$.

Q30.

Although many candidates were able to correctly write $\sqrt{x^3}$ in the form $x^{\frac{3}{2}}$ it was not uncommon to see the incorrect answers $k = 3.5$ or $k = \frac{1}{3}$. Although follow through marks were available, a significant minority of candidates subsequently differentiated $x^{\frac{1}{3}}$ incorrectly as $\frac{1}{3}x^{\frac{2}{3}}$. In addition to arithmetical errors the most common error in finding the equation of the tangent was to use the gradient of the tangent as $-\frac{1}{5}$; in effect, the equation of the normal was found. Very few candidates failed to write their equation in the

required form.

Q31.

Part (a) was well answered by the majority of candidates. Many fully correct responses were seen and if there were errors it was usually for failing to multiply by the derivative of $(3x + 1)$ so answers of $5(3x + 1)^4$ were not uncommon.

Part (b) was not as well answered as part (a) but many correct responses were seen. The main error was again to miss the factor of 3 and the very common incorrect answer was $1/(3x + 1)$.

In part (c) most candidates used the product rule successfully but a number of candidates believed that the differential of a product was the product of the differentials.

Candidates who made errors in part (b) were able to recover and earn 2 marks. A very common incorrect response was to simply multiply the answers to parts (a) and (b) resulting in $45(3x + 1)^3$. Another common error was to 'simplify' $\ln(3x + 1)(3x + 1)^4$ to obtain $\ln(3x + 1)^5$.

Q32.

Part (a) of this question was reasonably answered with most candidates obtaining the method mark for the product rule. Some candidates then lost the accuracy mark through incorrect evaluation of the constant associated with the derivative of $\cos 2x$; 2 and $\frac{1}{2}$ were frequently seen.

Answers to part (b)(i), although frequently correct, were often badly set out with the function being equated to zero or x being changed to α at various points in the solution. Division of $\cos 2x$ by $\cos 2x$ often resulted in zero before going on to fudge the correct answer. Part (b)(ii) was usually well answered with correct evaluations of $f(0.4)$ and $f(0.5)$. Some candidates then lost marks by just saying α was 'between these two values' without stipulating 0.4 and 0.5. Part (b)(iii) was usually well done but there were many cases when

the $\tan$ became removed from its $2x$ and $\frac{1}{2} x \tan^{-1} = x$ was often seen. Part (b)(iv) was generally well answered but marks were lost from the use of degrees and for not writing the answer to the required degree of accuracy.

Many fully correct answers and many answers which only lost the final accuracy mark were seen in part (c). Other answers which had the wrong coefficient associated with the $\sin 2x$ often got the method marks. There were candidates who started with $u = x$ and $v =$

$\cos 2x$ or started with $\frac{dv}{dx} = x$ and obtained a more difficult integral.

Q33.

Part (a) Candidates did not seem confident at working on this kind of problem and algebraic weaknesses were evident. Many worked backwards from the result in part (a)(i) and did not always convince the examiner that they were considering the surface area of four faces and the base. The inability of most candidates to rearrange the formula to make h the subject in part (a)(ii) was alarming. Consequently few, without considerable fudging,

could establish the printed formula for the volume. Part (b) Basic differentiation is well

understood and most candidates found $\frac{dV}{dx}$ correctly. Some tried to substitute $x = 3$ into the expression for V in order to show there was a stationary point, but usually this part was answered well.

Part (c) It was not uncommon to see the second derivative as $4x$ even though the first derivative was correct. A generous follow through was given here provided candidates could interpret the value of their second derivative.

Q34.

Part (a)(i) Some candidates ignored the request to state the coordinates of B even though they were using the height of the triangle as 5. The negative x -coordinate of A caused quite a few to feel that the triangle had a negative area. Far too many when finding

$\frac{1}{2} \times 1 \times 5$ wrote the answer as 3.

Part (a)(ii) Practically every candidate found the correct integral although some made errors when cancelling fractions.

Part (a)(iii) It was necessary here to have the lower limit as -1 and the upper limit as 0. Many reversed the order and by some trickery arrived at a positive value. This was penalised and so very few, even though many had an answer of 1 for the area, scored full marks for this part of the question.

Part (b) Most candidates differentiated correctly but, because of poor understanding of negative signs, many wrong values of -13 were seen for the gradient. There is obviously confusion for many between tangents and normals and several thought the gradient of the

tangent was $\frac{-1}{17}$.

Q35.

This calculus question again proved to be a good source of marks for many candidates.

Part (a)(i) was answered very well but in part (a)(ii) a significant minority used $\frac{d^2x}{dx^2} = 0$

instead of $\frac{dy}{dx} = 0$ dx^2 . In part (a)(iii), although most candidates showed that they understood the method to find the gradient of the tangent, a significant proportion failed to find and use the gradient of the normal. Many candidates showed that they had a good understanding of integration but, surprisingly, errors in integrating $(x + 1)$ were almost as

common as errors in integrating $\frac{4}{x^2}$. Examiners expected to see some simplification of

the algebraic term $\frac{+4x^{-1}}{-1}$. In the final part, the evaluation of $F(4) - F(1)$ as 8.5 was the most common error.